Dynamic Attention Scheduling for Multi-Subject Identity-Preserving Generation with Automatic Pipeline

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Abstract

Recent progress in multi-subject image generation has been driven by similarity-editing frameworks such as MUSE, which inject subject-specific features during diffusion to preserve identities. However, MUSE relies on a fixed attention injection and manual bounding-box layouts, both of which limit identity fidelity, spatial separation, and usability. In this work, we introduce ***dynamic attention scheduling*** *for MUSE, a novel mechanism that adaptively adjusts the injection strength of each subject's features across diffusion timesteps. By replacing the original static schedule with a subject-aware, time-varying formulation, our method significantly improves identity preservation and reduces cross-subject interference in complex multi-subject scenes. Complementing this, we propose an* ***LLM-guided layout predictor*** *that infers semantically meaningful bounding boxes directly from a natural-language prompt and reference images, fully eliminating the need for manual user input. Extensive experiments demonstrate that our approach produces higher identity consistency, improved spatial coherence, and more reliable multi-subject generation compared to MUSE, while requiring zero manual layout specification. Our results advance similarity-editing methods toward a practical, fully automatic multi-subject generation pipeline.*

1. Introduction

Multi-subject identity-preserving image generation has become an increasingly important problem in personalized content creation, virtual staging, and multi-character synthesis. Recent work such as MUSE [3] introduces a similarity-editing framework that injects subject-specific features into the diffusion process, enabling scenes composed of multiple reference subjects. While MUSE represents a strong step toward controllable multi-subject generation, it still exhibits several notable limitations. First, its fixed attention injection schedule often fails to maintain consistent identities: subjects may appear distorted, partially missing, or incorrectly blended with others. Second, MUSE frequently introduces unintended duplicate subjects, especially when multiple entities appear spatially close (e.g., generating two cats and one dog when only one of each is required)As shown in Figure 1 .Third, MUSE relies heavily on manually specified bounding boxes, significantly limiting usability for end users. To address these challenges, we propose a fully automated and more robust variant of MUSE centered around a novel **dynamic attention scheduling** mechanism. Instead of relying on MUSE's static, one-size-fits-all injection pattern, our dynamic schedule adaptively balances *general structural consistency* (e.g., correct subject placement and global composition) and *identity preservation* across diffusion timesteps. Early stages emphasize coarse structural fidelity, while later stages gradually increase the strength of identity-specific injections. This process substantially mitigates subject duplication, reduces cross-subject interference, and improves

Figure 1. MUSE incorrectly synthesizes an additional cat instance in the background, despite the prompt specifying only one cat and one dog.

identity preservation, yielding clear visual improvements over the original MUSE baseline. Dynamic scheduling is the core contribution of our work and directly targets the most critical weaknesses of existing similarity-editing pipelines.

Complementing our attention improvements, we introduce an **automatic layout generation pipeline** based on Qwen 2.5 VL (7B). Given a natural-language description of up to ten subjects, the LLM outputs structured bounding boxes in a strict JSON format, with normalized coordinates constrained to a $(1024, 1024)$ canvas. The system prompt enforces consistent formatting, one-decimal precision, and prohibits extraneous commentary. This component entirely removes MUSE's requirement for manual box annotation, substantially improving usability and enabling fully automated multi-subject generation.

Our key contributions are as follows:

- We propose a novel **dynamic attention scheduling** mechanism for similarity-editing diffusion models, improving identity preservation, mitigating subject duplication, and enhancing cross-subject separation. This mechanism replaces MUSE's rigid static schedule with an adaptive formulation that better balances structural and identity-level consistency across timesteps.
- We introduce an **automatic layout generation pipeline** using Qwen 2.5 VL (7B), which infers subject bounding boxes directly from natural-language descriptions, eliminating the need for manual user-specified layouts.
- Together, these contributions form a **fully automated multi-subject generation pipeline** that achieves superior qualitative results and is strongly preferred in user studies compared to the original MUSE framework.

2. Related Work

Multi-subject image customization is crucial for personalized content creation and multi-entity scene synthesis. Classical subject-driven methods such as DreamBooth and Textual Inversion mainly handle a single subject and struggle with multi-subject identity mixing and spatial inconsistency. Recent work tackles these issues from different angles.

MuDI (Identity Decoupling for Multi-Subject Personalization) [1] addresses identity bleeding by using segmentation-guided identity decoupling and Seg-Mix augmentation. Subject-aware masks are used at inference to keep identities separated. While effective at preventing cross-subject leakage, MuDI depends heavily on segmentation quality and requires additional training, which increases pipeline complexity.

MultiBooth [4] adopts a two-stage pipeline: each subject is first learned individually, then combined via region-aware cross-attention and optional bounding-box constraints. This yields explicit spatial control and more stable compositions, but requires multiple fine-tuning stages and manual region specification, making it hard to scale to many subjects.

FreeFuse [2] is a training-free alternative that derives subject masks from cross-attention maps at inference time and applies LoRA modules only to the corresponding regions. This avoids retraining and is easy to use, but its automatically inferred masks are less precise, especially under occlusion, leading to weaker identity preservation in challenging scenes.

In summary, MuDI offers strong identity separation but relies on segmentation and extra training, MultiBooth provides structured spatial control at the cost of heavy fine-tuning and manual effort, and FreeFuse is lightweight but less precise. These trade-offs motivate our approach, which aims to preserve identity and spatial coherence in multi-subject generation while remaining fully automatic.

3. Method

3.1. Cross-Attention in Diffusion Models

Diffusion models rely heavily on cross-attention layers to inject conditioning signals such as text or subject-specific features. Given a query matrix Q from the UNet latent features and key–value matrices (K, V) derived from the conditioning encoder (e.g., CLIP), a cross-attention update is computed as

$$\mathrm{CA} = \sigma\left(\frac{QK^\top}{\sqrt{d}}\right) V, \tag{1}$$

where $\sigma(\cdot)$ denotes the softmax operation over attention scores and d is the feature dimension. Intuitively, the attention weights $\sigma(QK^\top/\sqrt{d})$ determine how much each token in the conditioning input contributes to the latent update at each spatial location.

3.2. Decoupled Cross-Attention for Subject Injection

To enable subject-conditioned generation, MUSE extends the standard cross-attention in Eq. 1 by introducing an additional branch that injects features from reference images as shown in Figure 2. Concretely, let (K, V) denote the key and value projections of the original conditioning signal (e.g., CLIP-encoded text), and let (K_I, V_I) denote the key and value projections of the injected subject image features. The decoupled cross-attention update is defined as

$$\mathrm{DCA} = \sigma\left(\frac{QK^\top}{\sqrt{d}}\right) V + \lambda\,\sigma\left(\frac{QK_I^\top}{\sqrt{d}}\right) V_I, \tag{2}$$

where $\sigma(\cdot)$ is the softmax over attention scores, d is the feature dimension, and λ is a scalar strength factor that controls how strongly the injected subject features influence the latent update.

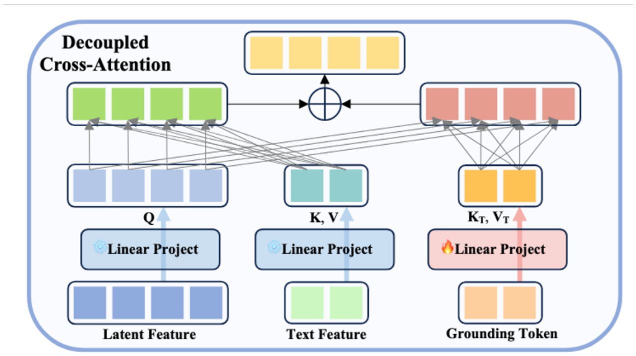

Figure 2. Decoupled cross-attention module used in MUSE [3]. The latent features provide the queries Q, while text features and grounding tokens are projected to keys and values (K, V) and (K_T, V_T), which are combined to update the latent representation. Figure adapted from MUSE.

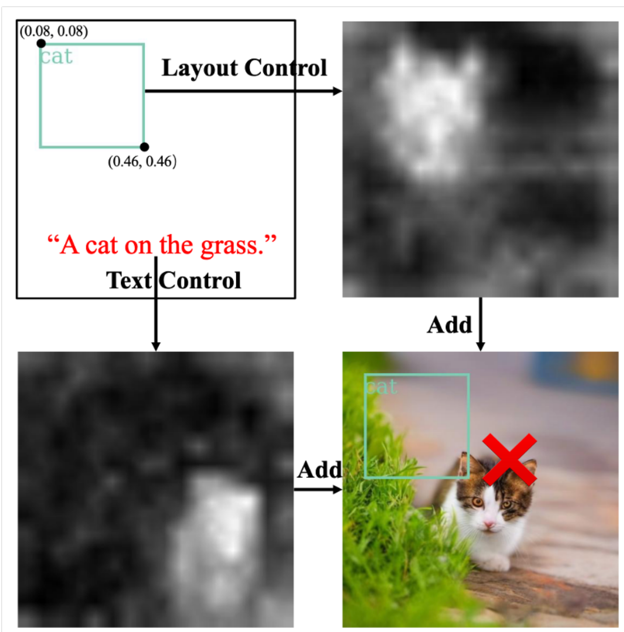

Figure 3. Control collision in the decoupled cross-attention (DCA) of MUSE [3]. Layout and text branches produce separate attention maps that are simply added, which can shift the effective focus away from the specified bounding box. Figure reproduced from MUSE [3].

The first term in Eq. 2 preserves the semantics and structure dictated by the original conditioning (e.g., the prompt), while the second term injects identity-specific information from the reference images. By decoupling these two contributions and modulating the injection strength via λ, MUSE can steer the generated image toward the desired subjects while still respecting the global scene description.

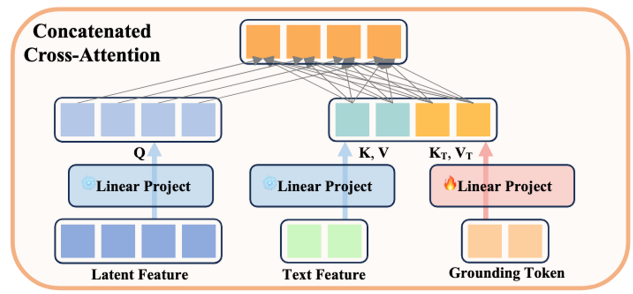

Figure 4. Concatenated cross-attention (CCA) in MUSE [3]. Text and grounding tokens are concatenated and attended jointly, yielding a single coherent attention distribution. Figure reproduced from MUSE [3].

3.2.1. Limitations of Decoupled Cross-Attention

The decoupled formulation in Eq. 2 separates the contribution of text and injected subject features, but layout and text control are still handled by independent branches whose attention maps are added. This *control collision* causes conflicting guidance over the same region, as seen in Figure 3: for the prompt “a cat on the grass”, the combined attention can place the cat outside the intended box, violating the layout constraint.

3.2.2. Concatenated Cross-Attention

To reduce control collision, MUSE [3] introduces *concatenated cross-attention* (CCA), which attends to text and layout tokens in a single operation. Let (K, V) be the text keys and values and (K_T, V_T) the grounding tokens; CCA is defined as

$$\mathrm{CCA} = \sigma\left(\frac{Q[K, K_T]^\top}{\sqrt{d}}\right)[V, V_T], \tag{3}$$

where $[K, K_T]$ and $[V, V_T]$ denote concatenation along the token axis (Figure 4). A single softmax over the joint token set lets the model resolve competition between text and layout cues within one coherent attention map.

3.2.3. Final Cross-Attention in MUSE

The final cross-attention (FCA) used by MUSE combines CCA for text–layout control with the subject-injection branch of DCA. Denoting the keys and values of the injected subject image features by (K_I, V_I), the FCA update can be written as

$$\mathrm{FCA} = \underbrace{\sigma\left(\frac{Q[K, K_T]^\top}{\sqrt{d}}\right)[V, V_T]}_{\text{text + layout control (CCA)}} + \lambda\, \underbrace{\sigma\left(\frac{QK_I^\top}{\sqrt{d}}\right)V_I}_{\text{subject injection}}, \tag{4}$$

where λ is a scalar strength factor controlling how strongly the injected subject features influence the latent update. In the original MUSE, λ is fixed across timesteps; in the next

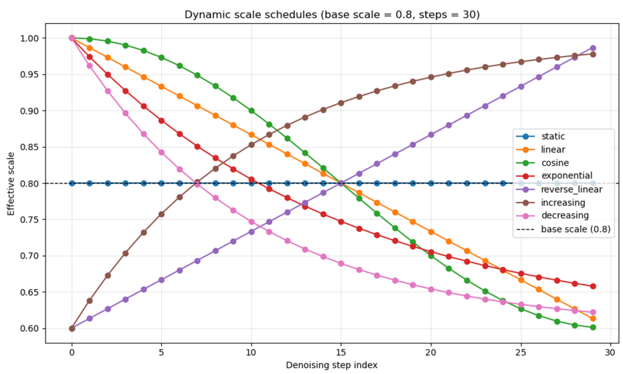

Figure 5. Dynamic attention schedules (example with $\lambda_{\text{base}} = 0.8$ and $T = 30$ steps). The static schedule keeps λ_t constant, while the increasing and decreasing schedules respectively strengthen or weaken subject attention over time. We primarily use the static, increasing, and decreasing variants.

subsection we replace this static coefficient with a time-varying *dynamic attention schedule* λ_t to better balance structure preservation and identity fidelity during the denoising process.

4. Dynamic Attention Scheduling

4.1. Time-Dependent Injection Schedules

In the original MUSE, the subject-injection strength λ in Eq. 4 is fixed across all diffusion timesteps. We instead use a time-dependent schedule λ_t that modulates how strongly reference features are injected at each step t:

$$\lambda_t = \lambda_{\text{base}}\, s\left(\frac{t}{T}\right), \tag{5}$$

where T is the total number of steps and $s(\cdot)$ is a shaping function over normalized time. Figure 5 visualizes several candidate schedules, including the static baseline and the increasing/decreasing schedules we focus on in this work.

4.2. Static vs. Increasing vs. Decreasing

We compare three schedules in practice: a *static* schedule (original MUSE), an *increasing* schedule that gradually strengthens subject attention, and a *decreasing* schedule that does the opposite. Figure 8 shows an example for the prompt “a hat on the cat”.

4.3. Advantage of Increasing Attention

Increasing Attention largely **remove hallucination of the diffusion model on the subject number**. As in Figure 1, the static attention accidentally add one more cat in the background. However, after we use the increase attention in the Figure 6 the number of cat is correct.

Figure 6. When using the increasing attention the unexpected cat was removed.

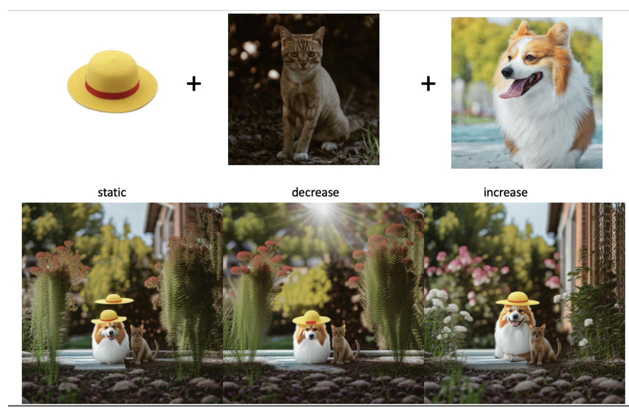

Figure 7. Increasing attention achieved a batter quality and spatial relationship. The prompt of those generation is ”A dog wearing the hat, setting close to cat, they are in garden”. the reference graph is the dog, the cat and the hat.

Additionally, the increasing attention can **achieve a batter spatial relationship**. As shown in the Figure 7, the diffusion model using increase attention not only have a batter result, but also have a more reasonable spatial relationship. The cat and the dog is sitting in the garden, static and decrease attention makes the flower too large comparing to the cat and dog.

The increasing attention

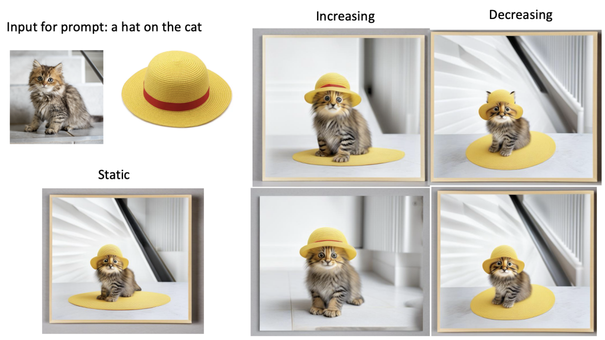

Figure 8. Comparison of static, increasing, and decreasing subject-attention schedules for the prompt “a hat on the cat”. The static schedule fails to preserve the identity of the hat (the red band disappears), and the decreasing schedule injects subject features too strongly in early, noisy steps, causing distortions. Our increasing schedule better preserves both the cat identity and the hat appearance, including the red band.

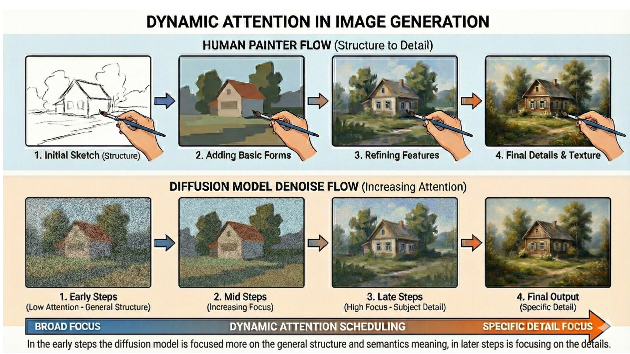

Figure 9. Intuition for dynamic attention scheduling (figure adapted from our illustration). Top: human painters proceed from global structure to fine detail. Bottom (conceptual): our increasing schedule follows the same principle by using low subject-attention strength in early diffusion steps (focusing on layout and semantics) and gradually increasing λ_t so that later steps refine identity-specific details.

4.4. Why Increasing Attention Works Better

The behavior of the increasing schedule can be understood by analogy to how human artists paint. As illustrated in Figure 11, a painter typically starts with a rough layout, then refines shapes, and finally adds high-frequency identity details.

Our increasing schedule is designed to mimic this process: early timesteps, when noise is high, use a small λ_t so that the model primarily establishes the global scene structure dictated by text and layout. As noise decreases, λ_t increases, allowing the injected subject features to dominate and refine details such as facial traits or the red band on the hat. This alignment between the denoising dynamics and the human painting workflow explains why the increasing schedule yields sharper, more faithful identities than the static or decreasing baselines.

4.5. User Study

We conducted a small user study to assess perceived quality. For three different prompts, we generated triplets of images using the static, increasing, and decreasing schedules (9 images per participant). Ten participants were asked, for each triplet, to choose the image that best matched the prompt while preserving both subject identity and layout. Table 1 reports statistics summarizing the choices.

Participants overwhelmingly preferred the images generated with the increasing schedule, supporting our qualitative observation that dynamic, increasing subject attention provides a better trade-off between structural coherence and identity fidelity than either a static or decreasing schedule.

Table 1. User study preference for different attention schedules (placeholder values). Each of the 10 participants evaluated 5 triplets (50 votes total).

Schedule	Votes (out of 30)	Preference (%)
Static	3	10
Increasing	25	83
Decreasing	2	6

5. Discussion

Our results show that a simple, inference-time modification of MUSE [3] can substantially improve multi-subject identity preservation and usability. By replacing the static subject-injection coefficient with an increasing schedule λ_t, we better align the denoising dynamics with the coarse-to-fine process used by human artists: early steps focus on global structure and semantics, while later steps refine identity-specific details. Qualitative comparisons, together with our small user study, consistently indicate that the increasing schedule produces images that users perceive as more faithful to both the prompt and the reference subjects than static or decreasing schedules.

The LLM-guided layout module complements this behavior by removing the need for manually specified bounding boxes. In practice, we find that Qwen 2.5 VL (7B) is capable of proposing reasonable layouts for a broad range of prompts, and that the resulting automatic boxes are often sufficient for high-quality generation. This suggests that multi-subject customization systems can move toward truly text-driven workflows, where users only describe the desired scene and supply reference images.

Our approach still has several limitations. First, the dynamic schedule is hand-designed rather than learned; while the increasing pattern works well across our experiments, a data-driven or per-scene adaptive schedule may further improve robustness.Third, because we keep both the base diffusion model and the large language modules active throughout the denoising process, our method slightly increases VRAM usage compared to the original MUSE pipeline, which may limit batch size on resource-constrained GPUs.

6. Appendix

Few more examples

Figure 10. Example of wearing 4 clothes for the model

Figure 11. Example of two dogs and one cat

References

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